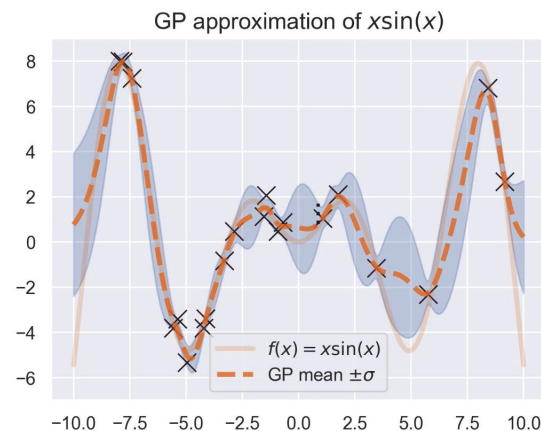


Bayesian Optimization and Active Learning Intro

Week 3

Bayesian Optimization

- Can optimize/recreate black-box/compute-heavy function
- Generally want to attempt to recreate true function from limited number of queries
- Many are based on Gaussian Processes (stochastic process dependant on priors)
- Bayesian optimization creates
 - a probabilistic surrogate model to approximate the actual function
 - An acquisition function measures the 'potential' of points so next point to be evaluated chosen
- Multiple types of acquisition functions, including Thompson scattering and Expected Improvement
 - Possible that it may get stuck in a local maxima/minima



Reference:

<https://www.miguelgondou.com/blogposts/2023-07-31/intro-to-bo/#an-introduction-to-gaussian-processes>

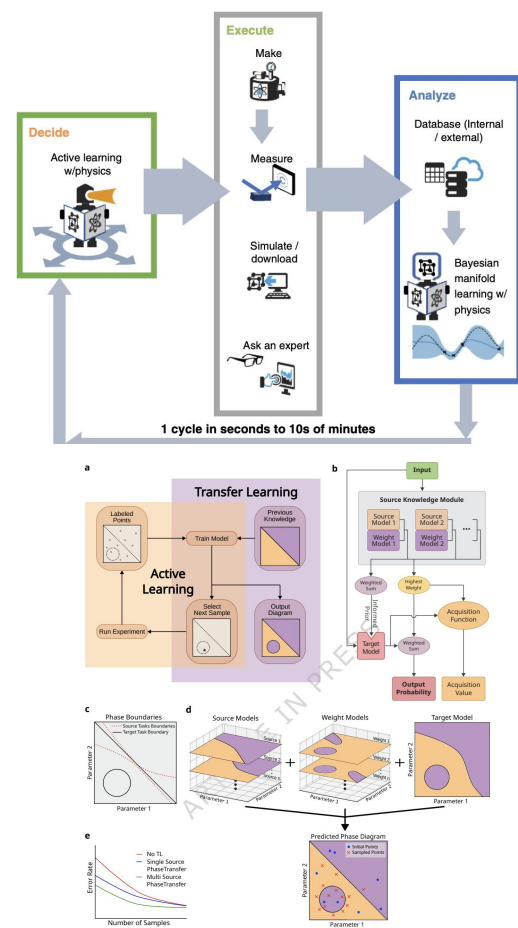
Active Learning

- Active learning is an ML method where the algorithm uses existing data to 'choose' new data to learn from
 - Useful for optimal experimental design to get desired knowledge with least amount of time
- New data taken is in a region with little confidence (possibly a boundary)
 - Existing data on a material or similar materials can also help inform what new data must be taken (transfer learning)

1. Select the most uncertain samples from the pool using the query strategy.
2. Query the labeler for the labels of the selected samples.
3. Add the labeled samples to the training set and remove them from the pool.
4. Update the model with the new training set.
5. Repeat until a stopping criterion is met.

Reference:

<https://gpttutorpro.com/active-learning-with-python-a-simple-example>



Reference: [On-the-fly Closed-loop Autonomous Materials Discovery via Bayesian Active Learning](#), Kusne, 2020

Code Example

```
1 # Define the number of iterations
2 n_iter = 10
3
4 # Run the active learning loop
5 for i in range(n_iter):
6     # Select the most uncertain samples from the pool
7     query_idx, query_inst = learner.query(pool)
8
9     # Query the labeler for the labels of the selected samples
10    query_labels = labeler.label(query_inst)
11
12    # Add the labeled samples to the training set and remove them from the pool
13    learner.teach(pool[query_idx], query_labels)
14    pool = np.delete(pool, query_idx, axis=0)
15
16    # Update the model with the new training set
17    learner.fit(X_train, y_train)
18
19    # Evaluate the performance of the model on the test set
20    acc = learner.score(X_test, y_test)
21
22    # Print the accuracy score
23    print(f"Iteration {i+1}: Accuracy = {acc:.3f}")
```

After running the code, you should see something like this:

```
Iteration 1: Accuracy = 0.800
Iteration 2: Accuracy = 0.867
Iteration 3: Accuracy = 0.933
Iteration 4: Accuracy = 0.933
Iteration 5: Accuracy = 0.933
Iteration 6: Accuracy = 0.933
Iteration 7: Accuracy = 0.933
Iteration 8: Accuracy = 0.933
Iteration 9: Accuracy = 0.933
Iteration 10: Accuracy = 0.933
```

Reference:

<https://gpttutorpro.com/active-learning-with-python-a-simple-example>

Active Learning and Synchrotrons

- Physics-informed Bayesian learning used to in neutron-diffraction experiment to determine magnetic ordering transition of two materials in 1/5th of time
- Transfer learning + active learning phase mapping results in 90% less data needed
- Autonomous control of x-ray scattering also has been used
- Important to make sure algorithm runs fast enough that no experimental time wasted waiting for processing

References:

[On-the-fly autonomous control of neutron diffraction via physics-informed Bayesian active learning](#), McDannald, 2022
[PhaseTransfer: A transfer learning framework for efficient phase diagram mapping](#), González-García, 2026
[Autonomous x-ray scattering](#), Yager et. al, 2023

Experiment	Input space dimension	Benefit
Autonomous X-ray scattering mapping ^{8,9,11}	2	Surveys parameter-space landscape and identifies regions of interest quickly, with beam utilization vastly increased (a sixfold reduction in the number of measurements required to reach a similar model quality to a grid scan ⁸)
Autonomous synchrotron infrared mapping ⁴⁹	2	Identifies regions of interest quickly without human biases, with up to 50 times fewer data points needed for interpretation, beam utilization increased and shorter timescale (higher time resolution) required, and observation of transient chemical processes enabled ^{47,49}
Autonomous discovery of correlated electron phases	2	Dataset size decreased to <10% of original size
Inelastic neutron scattering	4	No human intervention required, experiment time decreased from several days to one night, and hit rate inside the regions of interest is about twice as high as random and grid-based measurement selection

Reference:

[Gaussian processes for autonomous data acquisition at large-scale synchrotron and neutron facilities](#), Noack et. al, 2021